Submission 1: Vitamin D and CYP24A1 expression

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Research focus

This exploratory project describes early-pregnancy CYP24A1 expression in relation to baseline 25-hydroxyvitamin D, subsequent preeclampsia outcome and a derived vitamin D group. CYP24A1 is related to vitamin D metabolism. The threshold of 30 ng/mL follows the source study; it is not a universal diagnostic threshold. The continuous variable is `mat_vitD`. The categorical variables are `preg_preeclampsia` and `vitD_group`, the latter derived from `mat_vitD`.

● Step1 : Global settings and packages

The expression columns and covariate rows have different input orders. All matching below uses sample identifiers. All 157 samples are retained; no covariate or supplied expression value is missing.

```
# Load packages for data manipulation and visualization.
library(ggplot2)
library(dplyr)
```

● Step2 : Import raw data

```
# Import gene expression data and use the first column as row names.

cov_path <- "E:/桌面/raw/GSE85307_covar-1.csv"
expr_path <- "E:/桌面/raw/GSE85307_geneExpression.csv"
gene_path <- "E:/桌面/raw/GSE85307_geneNames.csv"

# Read input files from the project's data/raw folder - github
#cov_path <- "data/raw/GSE85307_covar-1.csv"
#expr_path <- "data/raw/GSE85307_geneExpression.csv"
#gene_path <- "data/raw/GSE85307_geneNames.csv"
```

```r
# Check file existence: TRUE means found; FALSE means not found.
file.exists(cov_path)
```

```
#> [1] TRUE
```

```r
file.exists(expr_path)
```

```
#> [1] TRUE
```

```r
file.exists(gene_path)
```

```
#> [1] TRUE
```

```r
# Import participant characteristics.
covar_raw <- read.csv(
  cov_path,
  stringsAsFactors = FALSE,
  check.names = FALSE
)

# Import gene expression data, using the first column as row names.
expr_raw <- read.csv(
  expr_path,
  row.names = 1,
  check.names = FALSE
)
# Import the probe-to-gene mapping table.
link_raw <- read.csv(
  gene_path,
  stringsAsFactors = FALSE,
  check.names = FALSE
)

# Check dimensions: number of rows followed by number of columns.
dim(covar_raw)
```

```
#> [1] 157  10
```

```r
dim(expr_raw)
```

```
#> [1] 500 157
```

```r
dim(link_raw)
```

```
#> [1] 750   2
```

Step3 : Clean covariates

```
# Define a function to remove the first label and trim whitespace.
strip_label <- function(x) trimws(sub("^[^:]*:\\s*", "", x, perl = TRU
E))

# Create a working copy and retain the raw data object.
clean <- covar_raw

# Remove labels from variables other than participant identifiers.
for (v in setdiff(names(clean), c("subjectID", "geoID"))) {
  clean[[v]] <- strip_label(clean[[v]])
}



# Identify variables that should be numeric.
num_vars <- c("mat_age_yrs", "delivery_gestAge_wks",
              "sampleCollection_gestAge_wks", "mat_BMI", "mat_vitD")



# Convert each variable and check for newly introduced missing values.
for (v in num_vars) {
  # Retain the values before conversion.
  before <- clean[[v]]
  # Convert text to numeric values.
  clean[[v]] <- suppressWarnings(as.numeric(before))
   # Stop if conversion turns a previously nonmissing value into NA.
  if (any(is.na(clean[[v]]) & !is.na(before))) stop("Numeric conversion
 failed: ", v)
}
```

Step4 :Validate identifiers and match samples

```
# Check identifiers, probe IDs, and gene names for duplicates.
stopifnot(!anyDuplicated(clean$geoID), !anyDuplicated(clean$subjectID),
          !anyDuplicated(colnames(expr_raw)), !anyDuplicated(rownames(e
xpr_raw)),
          !anyDuplicated(link_raw$PROBEID), !anyDuplicated(link_raw$GEN
E),
          # Check whether both files contain the same set of samples.
          setequal(clean$geoID, colnames(expr_raw)),
          # Check that categorical variables contain only expected valu
es.
          all(clean$preg_preeclampsia %in% c("Normal", "Preeclampsia")),
          all(clean$mat_asthma %in% c("0", "1")))
```

```r
# Convert expression data to a matrix.
expr <- as.matrix(expr_raw)
# Check that all expression values are numeric and finite.
stopifnot(is.numeric(expr), all(is.finite(expr)))

# Reorder expression columns to match the covariate sample order.
expr <- expr[, match(clean$geoID, colnames(expr)), drop = FALSE]
# Confirm exact agreement of sample identifiers and order.
stopifnot(identical(colnames(expr), clean$geoID))
# Reorder the mapping table to match expression probe order.
map <- link_raw[match(rownames(expr), as.character(link_raw$PROBEID)), ]

# Check that each probe has a valid matching gene name.
stopifnot(!anyNA(map$GENE), identical(as.character(map$PROBEID), rownam
es(expr)))

# Replace probe row names with gene names.
rownames(expr) <- map$GENE
```

● Step5 : Create groups and combine data

```r
# Define categorical variables and their display order.
clean <- clean %>% mutate(
  # Pregnancy outcome: display Normal first.
  preg_preeclampsia = factor(preg_preeclampsia, levels = c("Normal", "P
reeclampsia")),
  # Group vitamin D using the source study's 30 ng/mL threshold.
  vitD_group = factor(ifelse(mat_vitD < 30, "<30 ng/mL", ">=30 ng/mL"),
                      levels = c("<30 ng/mL", ">=30 ng/mL")),
  # Convert asthma codes to readable labels.
  mat_asthma = factor(mat_asthma, levels = c("0", "1"), labels = c("No",
 "Yes")),
  # Treat microarray batch as a categorical variable.
  batch_Microarray = factor(batch_Microarray))
# Transpose expression data to one sample per row and one gene per colu
mn.
all_genes <- as.data.frame(t(expr), check.names = FALSE)
# Combine covariates and expression after verifying sample order.
analysis_data <- bind_cols(clean, all_genes)
# Check the target gene and expected sample size.
stopifnot(
  "CYP24A1" %in% rownames(expr),
  nrow(analysis_data) == 157L
)
```

● Step6 : Print concise data checks

```r
# Count missing values in each covariate.
missing_table <- data.frame(variable = names(clean), missing = colSums
(is.na(clean)))



# Report sample size, expression rows, and outcome group counts.
data.frame(
  participants = nrow(clean),
  probes = nrow(expr),
  normal = sum(clean$preg_preeclampsia == "Normal"),
  preeclampsia = sum(clean$preg_preeclampsia == "Preeclampsia"),
  missing_covariate_values = sum(missing_table$missing)
)

#>   participants probes normal preeclampsia missing_covariate_values
#> 1          157    500    110           47                         0

# Count participants in each outcome and vitamin D subgroup.
group_counts <- analysis_data %>%
  count(
    preg_preeclampsia,
    vitD_group,
    .drop = FALSE
  )

# Print subgroup counts without printing the full dataset.
group_counts

#>   preg_preeclampsia vitD_group  n
#> 1            Normal  <30 ng/mL 78
#> 2            Normal >=30 ng/mL 32
#> 3      Preeclampsia  <30 ng/mL 41
#> 4      Preeclampsia >=30 ng/mL  6

# Verify that no covariate values are missing.
stopifnot(
  sum(missing_table$missing) == 0
)
```

The missing-value check passed, confirming that none of the covariate values in the cleaned dataset were missing. Therefore, no participants needed to be excluded because of missing covariate data.

➢ Figure 1: Expression distribution

```
# Plot a histogram of CYP24A1 expression.
p_hist <- ggplot(analysis_data, aes(CYP24A1)) +

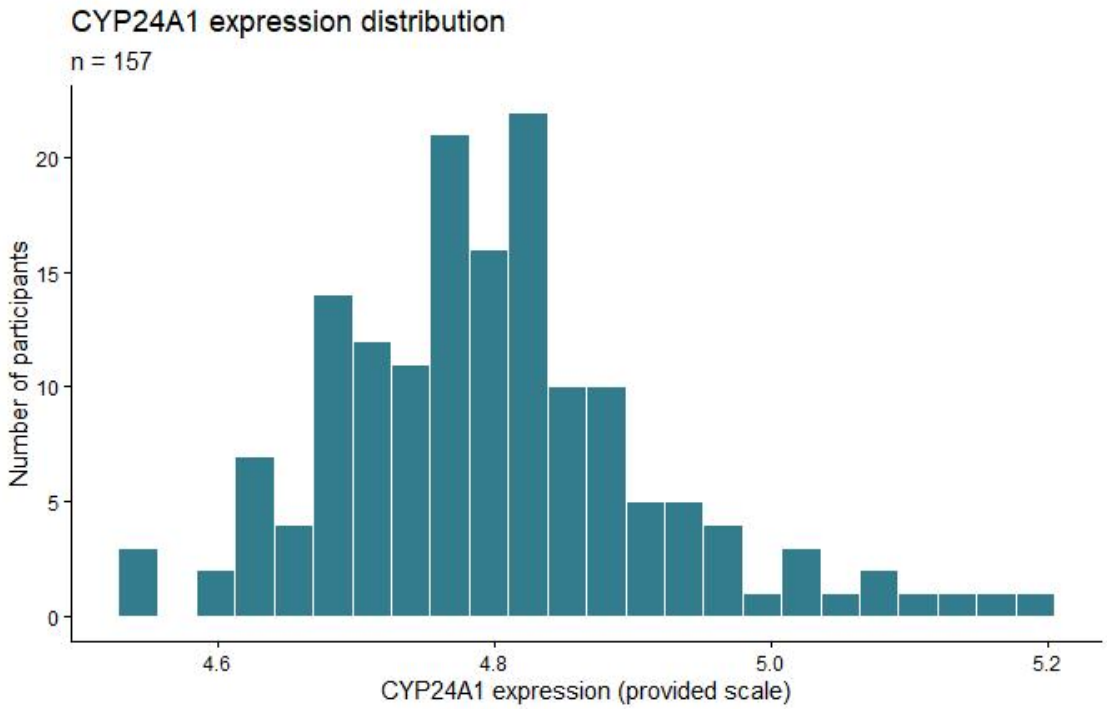
  # Use 24 bins; bar heights represent participant counts.

  geom_histogram(bins = 24, fill = "#337C8E", colour = "white") +

  # Add a title, dynamic sample size, and axis labels.

  labs(title = "CYP24A1 expression distribution", subtitle = "n = 157",
       x = "CYP24A1 expression (provided scale)", y = "Number of partic
ipants") +
  theme_classic(base_size = 11)

# Display the plot.
p_hist
```



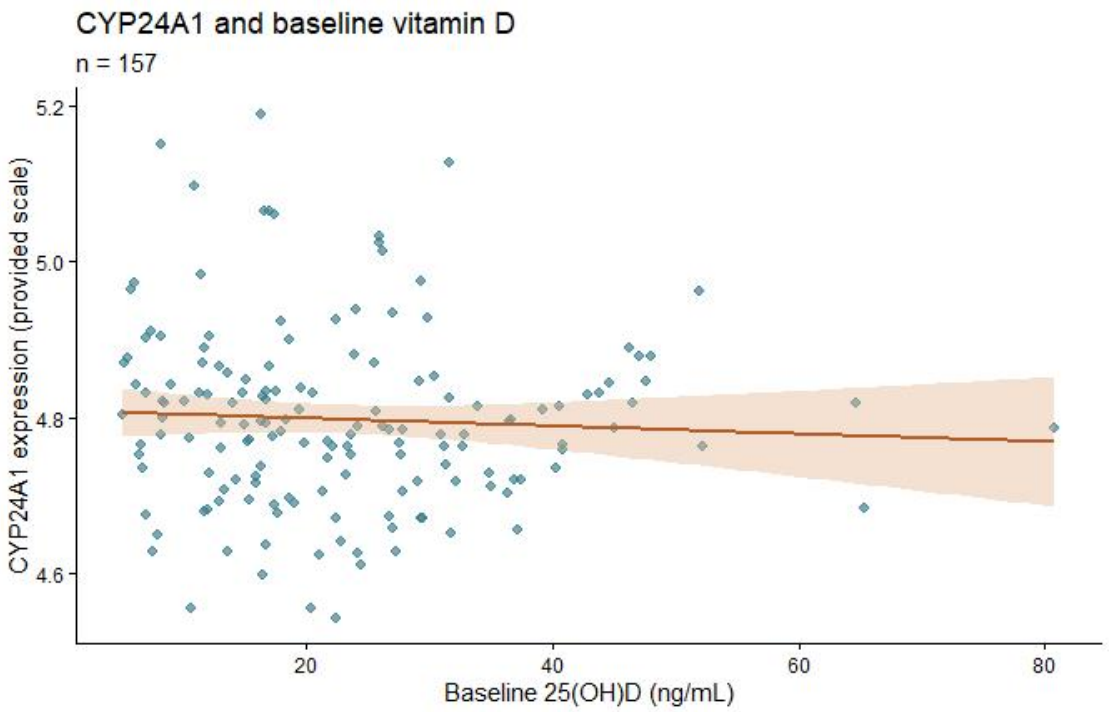
Interpretation figure1

Among the 157 participants, CYP24A1 expression is approximately unimodal and centered around 4.8 on the supplied expression scale. The distribution shows slight right skewness, with a few relatively high values extending to approximately 5.2. Most observations fall between approximately 4.6 and 5.0. The histogram alone is insufficient to establish normality.

➢ Figure 2: Continuous vitamin D and expression

```
# Map continuous vitamin D to x and CYP24A1 expression to y.
p_scatter <- ggplot(analysis_data, aes(mat_vitD, CYP24A1)) +
  # Each point represents one participant; transparency reduces overlap.
  geom_point(colour = "#337C8E", alpha = .65, size = 1.7) +
  # Add an unadjusted linear fit and a 95% confidence band for the mean.
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE,
              colour = "#B65E27", fill = "#E4B48B", linewidth = .8) +
  # Set the title and axis labels
  labs(title = "CYP24A1 and baseline vitamin D", subtitle = "n = 157",
       x = "Baseline 25(OH)D (ng/mL)", y = "CYP24A1 expression (provide
d scale)") +
  # Use the same theme as Figure 1.
  theme_classic(base_size = 11)

# Display the plot.
p_scatter
```



Interpretation - figure 2

The scatterplot shows little apparent linear association between baseline 25(OH)D and CYP24A1 expression. The unadjusted fitted line has a slight downward slope, but expression values vary considerably at similar vitamin D levels. The 95% confidence band widens at higher vitamin D levels, where observations are sparse,

indicating greater uncertainty in the fitted mean. This plot does not establish a causal relationship or rule out nonlinear associations.

➢ Figure 3: Both categorical covariates

```
# Count participants in each subgroup for plot annotations.
counts <- analysis_data %>% count(preg_preeclampsia, vitD_group)

# Position count labels above the highest expression value.
counts$label_y <- max(analysis_data$CYP24A1) + .05

# Map outcome to x, expression to y, and vitamin D group to fill.

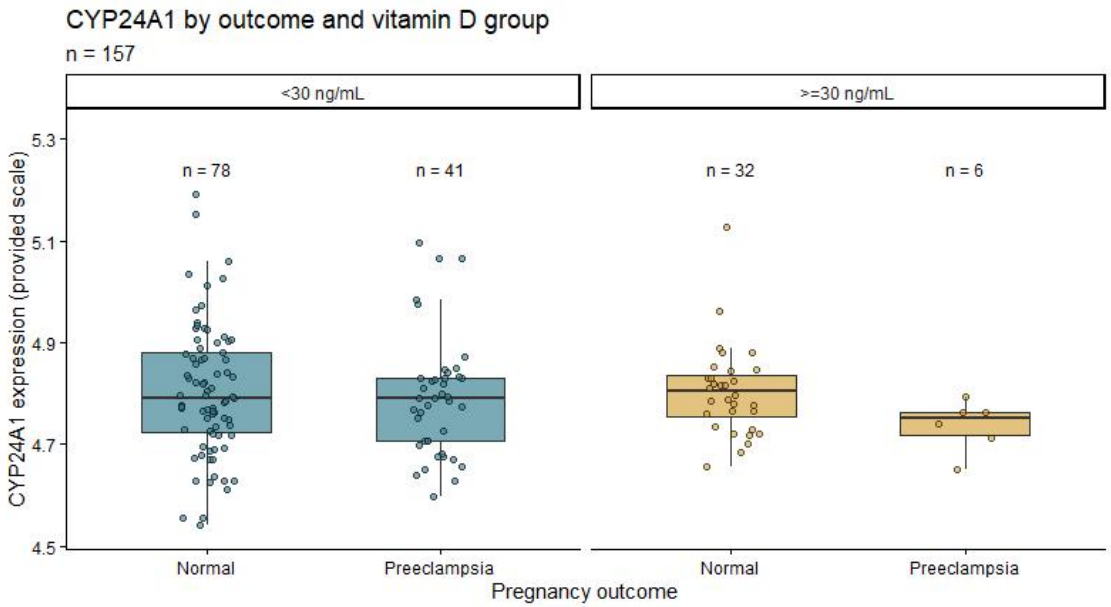
p_box <- ggplot(analysis_data, aes(preg_preeclampsia, CYP24A1, fill = v
itD_group)) +
  # Draw boxplots; individual points are shown in the next layer.

  geom_boxplot(width = .55, outlier.shape = NA, alpha = .65) +
  # Show each participant with horizontal jitter only.
  geom_point(position = position_jitter(width = .12, height = 0, seed =
 103),
             shape = 21, size = 1.3, alpha = .6) +

  # Display the sample size above each subgroup.

  geom_text(data = counts, aes(x = preg_preeclampsia, y = label_y,
                               label = paste0("n = ", n)),
            inherit.aes = FALSE, size = 3.1) +
  # Create one panel for each vitamin D group.
  facet_wrap(~vitD_group) +
  # Specify fill colors for the two groups.
  scale_fill_manual(values = c("#337C8E", "#D5A13B")) +
  # Add space for points and count labels.
  scale_y_continuous(expand = expansion(mult = c(.07, .17))) +

  # Set the title and axis labels.
  labs(title = "CYP24A1 by outcome and vitamin D group", subtitle = "n
= 157",
       x = "Pregnancy outcome", y = "CYP24A1 expression (provided scale)
") +
  # Keep a consistent theme and omit the redundant legend.
  theme_classic(base_size = 11) +
  theme(legend.position = "none")
# Display the plot.
p_box
```



CYP24A1 expression by subsequent pregnancy outcome and vitamin D group. Boxes show the median and interquartile range; Whiskers extend to the most extreme observed values within 1.5 times the IQR below the first quartile and above the third quartile. Jitter changes only horizontal display positions.

Interpretation figure3

In the <30 ng/mL vitamin D group, participants with normal pregnancy outcomes (n = 78) and those who developed preeclampsia (n = 41) have similar median CYP24A1 expression and substantially overlapping distributions. In the ≥30 ng/mL group, participants who developed preeclampsia (n = 6) have a slightly lower observed median and a narrower interquartile range than those with normal outcomes (n = 32). However, this comparison should be interpreted cautiously because the preeclampsia subgroup contains only six participants. These descriptive differences do not establish statistical significance or causality.

● Conclusion

Overall, CYP24A1 expression shows little apparent linear association with baseline vitamin D. Expression distributions overlap across pregnancy outcome groups, although the small preeclampsia subgroup in the higher vitamin D category limits interpretation. These findings are exploratory.

Submission notes and references

I used ChatGPT/Codex to help organize the R Markdown workflow and explain the code. I also watched YouTube videos on GitHub operations and project data processing to improve my understanding.

- Mirzakhani H, et al. Early pregnancy vitamin D status and risk of preeclampsia. J Clin Invest. 2016;126(12):4702–4715. doi:10.1172/JCI89031.
- GEO GSE85307: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE85307
- NCBI Gene CYP24A1: https://www.ncbi.nlm.nih.gov/gene/1591